

MATH-2307: Mathematics-III (Matrices, Linear System of Equations and Vector Analysis)

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Physical quantities can be divided into two main groups, scalar quantities and vector quantities:

Scalar Quantities

A Physical Quantity which has magnitude only is called as a Scalar. Example: Time, Temperature, Mass, Density, Volume, Energy, Distance, Speed, Specific Heat etc. are examples of scalars. That is, the measurement of years, months, weeks, days, hours, minutes, seconds, and even milliseconds, A temperature of 15°C, A mass of 0.2 kg, etc.

Vector Quantities

A Physical Quantity which has both magnitude and direction is called as Vector Examples: velocity, displacement, acceleration, force, Weight, Momentum, Magnetic Field Intensity etc.

Some Examples:

01. A speed of 10 km/h is a scalar quantity, but a velocity of 10 km/h due north is a vector quantity.
02. A temperature of 1000 c is a scalar quantity.
03. The weight of a 7 kg mass is a vector quantity. [$w = mg$]

Magnitude and Sign Convention of Vectors

Two dimensional vector is

$$\overrightarrow{OA} \text{ or } \vec{A} = A_x \hat{i} + A_y \hat{j} = \begin{bmatrix} A_x \\ A_y \end{bmatrix}$$

Length or magnitude is $|\overrightarrow{OA}|$ or $|\vec{A}| = \sqrt{A_x^2 + A_y^2}$

Three dimensional vector is

$$\overrightarrow{OA} \text{ or } \vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k} = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

Length or magnitude is $|\overrightarrow{OA}|$ or $|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$

Example

Determine the vector having initial point $P(1, 2, 3)$ and terminal point $Q(5, 3, -1)$ and find its magnitude.

Solution

$$\overrightarrow{PQ} = (5 - 1)\hat{i} + (3 - 2)\hat{j} + (-1 - 3)\hat{k} = 4\hat{i} + \hat{j} - 4\hat{k}$$

$$|\overrightarrow{PQ}| = \sqrt{4^2 + 1^2 + (-4)^2} = \sqrt{21}$$

Physical Significance of The scalar or dot product: Work done

We know

Work = Force applied $\times$ displacement happened

$$W = \vec{F} \cdot \vec{S}$$

Example

A block of mass m moves from point A to B along a smooth plane surface under the action of force as shown in the figure. Find the work done.

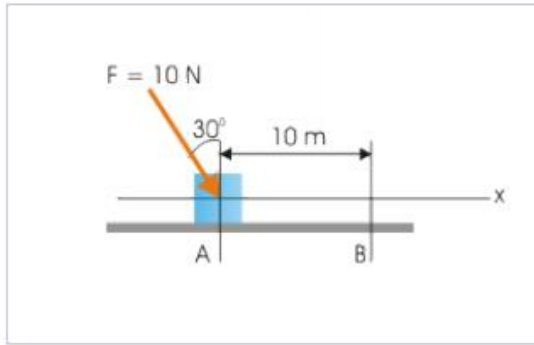


Figure 31

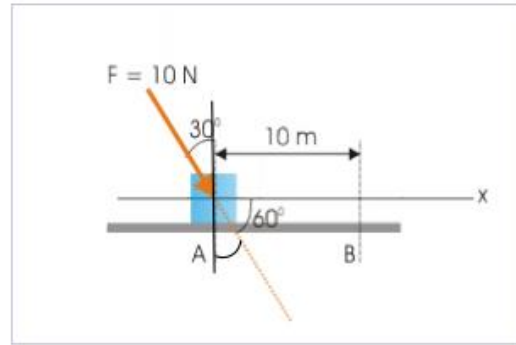
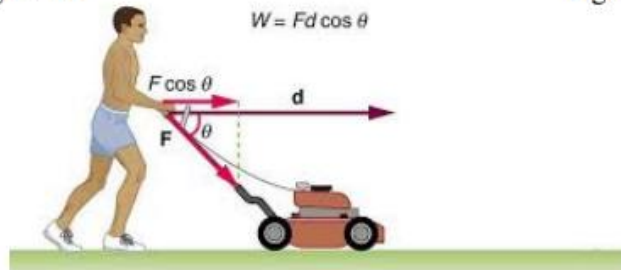


Figure 32



Solution

Workdone $W = \vec{F} \cdot \vec{S} = |\vec{F}| |\vec{S}| \cos \theta = 10 \cdot 10 \cdot \cos 60^\circ = 10 \cdot 10 \cdot \frac{1}{2} = 50$ joule

Dot Product of Two Vectors

If $\vec{A}$ and $\vec{B}$ are two vectors, then the dot product is defined by

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

If $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$, then $\vec{A} \cdot \vec{B}$ can be found by

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Note:

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1 \text{ and } \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{i} = \hat{i} \cdot \hat{k} = \hat{k} \cdot \hat{i} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{j} = 0$$

Example

If $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$, then show that

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Solution

Given vectors are $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$

$$\begin{aligned}
\vec{A} \cdot \vec{B} &= A_x B_x (\hat{i} \cdot \hat{i}) + A_x B_y (\hat{i} \cdot \hat{j}) + A_x B_z (\hat{i} \cdot \hat{k}) + A_y B_x (\hat{j} \cdot \hat{i}) + A_y B_y (\hat{j} \cdot \hat{j}) + A_y B_z (\hat{j} \cdot \hat{k}) \\
&\quad + A_z B_x (\hat{k} \cdot \hat{i}) + A_z B_y (\hat{k} \cdot \hat{j}) + A_z B_z (\hat{k} \cdot \hat{k}) \\
&= A_x B_x + A_y B_y + A_z B_z
\end{aligned}$$

Orthogonal (Perpendicular) Vectors

When two vectors $\vec{A}$ and $\vec{B}$ are perpendicular to each other, their dot product is always zero, that is

$$\vec{A} \cdot \vec{B} = 0$$

Since

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta = |\vec{A}| |\vec{B}| \cos 90^\circ = 0$$

Example

Determine whether the vectors $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} - 2\hat{k}$ are perpendicular.

Solution

Given that $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} - 2\hat{k}$

Now $\vec{A} \cdot \vec{B} = (3\hat{i} + 5\hat{j} - 2\hat{k}) \cdot (2\hat{i} - 2\hat{j} - 2\hat{k}) = 3 \cdot 2 + 5 \cdot (-2) + (-2) \cdot (-2) = 0$

Since $\vec{A} \cdot \vec{B} = 0$, the vectors $\vec{A}$ and $\vec{B}$ are perpendicular to each other.

Cross Product of Two Vectors

If $\vec{A}$ and $\vec{B}$ are two vectors, then the dot product is defined by

$$\vec{A} \times \vec{B} = \hat{n} |\vec{A}| |\vec{B}| \sin \theta$$

where $\hat{n}$ is the unit vector which will indicate the direction of $\vec{A} \times \vec{B}$.

If $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$, then $\vec{A} \times \vec{B}$ can be found by

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Example

If $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$, then show that

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Solution

Given vectors are $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$

$$\begin{aligned} \vec{A} \times \vec{B} &= A_x B_x (\hat{i} \times \hat{i}) + A_x B_y (\hat{i} \times \hat{j}) + A_x B_z (\hat{i} \times \hat{k}) + A_y B_x (\hat{j} \times \hat{i}) + A_y B_y (\hat{j} \times \hat{j}) \\ &\quad + A_y B_z (\hat{j} \times \hat{k}) + A_z B_x (\hat{k} \times \hat{i}) + A_z B_y (\hat{k} \times \hat{j}) + A_z B_z (\hat{k} \times \hat{k}) \\ &= A_x B_x \cdot 0 + A_x B_y \hat{k} + A_x B_z (-\hat{j}) + A_y B_x (-\hat{k}) + A_y B_y \cdot 0 + A_y B_z \hat{i} + A_z B_x \hat{j} + \\ &\quad A_z B_y (-\hat{i}) + A_z B_z \cdot 0 \\ &= \hat{i}(A_y B_z - A_z B_y) + \hat{j}(A_x B_z - A_z B_x) + \hat{k}(A_x B_y - A_y B_x) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \end{aligned}$$

Parallel Vectors

When two vectors $\vec{A}$ and $\vec{B}$ are parallel to each other, their cross product is always zero, that is

$$\vec{A} \times \vec{B} = 0$$

Since

$$\vec{A} \times \vec{B} = \hat{n}|\vec{A}||\vec{B}|\sin\theta = \hat{n}|\vec{A}||\vec{B}|\sin 0^\circ = 0$$

Example

Determine whether the vectors $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} - 2\hat{k}$ are parallel.

Solution

Given that $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} - 2\hat{k}$

$$\text{Now } \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 5 & -2 \\ 2 & -2 & -2 \end{vmatrix} \\
&= (-10 - 4)\hat{i} + (-4 + 6)\hat{j} + (-6 - 10)\hat{k} \\
&= -14\hat{i} + 2\hat{j} - 16\hat{k} \\
&\neq 0
\end{aligned}$$

Hence the given vectors are not mutually parallel.

Example

Find the angle between the vectors $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{B} = 4\hat{i} + \hat{j} - 3\hat{k}$.

Solution

Given that $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{B} = 4\hat{i} + \hat{j} - 3\hat{k}$. Now

$$\begin{aligned}
\vec{A} \cdot \vec{B} &= |\vec{A}||\vec{B}| \cos \theta \\
\cos \theta &= \frac{\vec{A} \cdot \vec{B}}{|\vec{A}||\vec{B}|} \\
\cos \theta &= \frac{2 \cdot 4 + (-3) \cdot 1 + 1 \cdot (-3)}{\sqrt{2^2 + (-3)^2 + 1^2} \sqrt{4^2 + 1^2 + (-3)^2}} \\
\cos \theta &= \frac{2}{\sqrt{14}\sqrt{26}} \\
\theta &= \cos^{-1} \left(\frac{2}{\sqrt{14}\sqrt{26}} \right)
\end{aligned}$$

Example

Determine the angles α, β, γ which the vector $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$ makes with the positive directions of the coordinate axes and show that $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$.

Solution

Given vector is $\vec{A} = 2\hat{i} - 3\hat{j} + \hat{k}$

Unit vector along the x -axis is $\hat{i}$

Thus, the angle α making by the vector $\vec{A}$ with the x -axis is given by

$$\begin{aligned}
 \alpha &= \cos^{-1} \left(\frac{\vec{A} \cdot \hat{i}}{|\vec{A}| |\hat{i}|} \right) \\
 &= \cos^{-1} \left(\frac{2.1 + (-3) \cdot 0 + 1 \cdot 0}{\sqrt{2^2 + (-3)^2 + 1^2} \cdot \sqrt{1^2 + 0^2 + 0^2}} \right) \\
 &= \cos^{-1} \left(\frac{2}{\sqrt{14}} \right)
 \end{aligned}$$

Unit vector along the y -axis is $\hat{j}$

Thus, the angle β making by the vector $\vec{A}$ with the x -axis is given by

$$\begin{aligned}
 \beta &= \cos^{-1} \left(\frac{\vec{A} \cdot \hat{j}}{|\vec{A}| |\hat{j}|} \right) \\
 &= \cos^{-1} \left(\frac{2.0 + (-3) \cdot 1 + 1 \cdot 0}{\sqrt{2^2 + (-3)^2 + 1^2} \cdot \sqrt{0^2 + 1^2 + 0^2}} \right) \\
 &= \cos^{-1} \left(-\frac{3}{\sqrt{14}} \right)
 \end{aligned}$$

Unit vector along the z -axis is $\hat{k}$

Thus, the angle γ making by the vector $\vec{A}$ with the x -axis is given by

$$\begin{aligned}
 \gamma &= \cos^{-1} \left(\frac{\vec{A} \cdot \hat{k}}{|\vec{A}| |\hat{k}|} \right) \\
 &= \cos^{-1} \left(\frac{2.0 + (-3) \cdot 0 + 1 \cdot 1}{\sqrt{2^2 + (-3)^2 + 1^2} \cdot \sqrt{1^2 + 0^2 + 0^2}} \right) \\
 &= \cos^{-1} \left(\frac{1}{\sqrt{14}} \right)
 \end{aligned}$$

Example

A particle acted on by constant forces $\vec{F}_1 = 4\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{F}_2 = 3\hat{i} + \hat{j} - \hat{k}$ (both measured in Newton), is displaced from the point (1, 2, 3) to the point (5, 4, 1) (measured in meters). Find the total work done by the forces.

Solution

fsajh

Scalar triple product

If $A = [a_{ij}]$ where $a_{ij} = \begin{cases} 0, & \text{when } i < j \\ i + j, & \text{when } i = j \\ 2i - j, & \text{when } i > j \end{cases}$

Example

Find a unit vector in the direction to the vector $\vec{a} = 2\hat{i} + 4\hat{j} - 5\hat{k}$.

Solution

Given vector is $\vec{a} = 2\hat{i} + 4\hat{j} - 5\hat{k}$

The magnitude of the vector $\vec{a}$ is

$$|\vec{a}| = \sqrt{2^2 + 4^2 + (-5)^2} = \sqrt{45}$$

Thus, the unit vector parallel to the vector $\vec{a}$ is

$$\hat{e} = \frac{2\hat{i} + 4\hat{j} - 5\hat{k}}{\sqrt{45}} = \frac{2}{\sqrt{45}}\hat{i} + \frac{4}{\sqrt{45}}\hat{j} - \frac{5}{\sqrt{45}}\hat{k}$$

Example

Find a unit vector parallel to the resultant of vectors $\vec{a} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.

Solution

the resultant of vectors $\vec{a} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$ is given by

$$\begin{aligned} \vec{R} &= \vec{a} + \vec{b} \\ &= (2 + 1)\hat{i} + (4 + 2)\hat{j} + (-5 + 3)\hat{k} \\ &= 3\hat{i} + 6\hat{j} - 2\hat{k} \end{aligned}$$

Magnitude of the resultant, $|\vec{R}| = \sqrt{3^2 + 6^2 + (-2)^2} = \sqrt{49} = 7$

Thus, the unit vector parallel to the resultant of vectors $\vec{a}$ and $\vec{b}$ is

$$\hat{e} = \frac{\vec{R}}{|\vec{R}|} = \frac{3\hat{i} + 6\hat{j} - 2\hat{k}}{7} = \frac{3}{7}\hat{i} + \frac{6}{7}\hat{j} - \frac{2}{7}\hat{k}$$

Example

Find a unit vector perpendicular to the vectors $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 2\hat{k}$.

Solution

A vector perpendicular to $\vec{a}$ and $\vec{b}$ is $\vec{a} \times \vec{b}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 0 \\ -1 & 2 & 2 \end{vmatrix} = \hat{i}(2 - 0) - \hat{j}(6 - 0) + \hat{k}(6 + 1) = 2\hat{i} - 6\hat{j} + 7\hat{k}$$

Now, a unit vector perpendicular to $\vec{a}$ and $\vec{b}$ is obtained by dividing the vector $\vec{a} \times \vec{b}$ by its magnitude.

$$\therefore \hat{e} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{2\hat{i} - 6\hat{j} + 7\hat{k}}{\sqrt{2^2 + (-6)^2 + 7^2}} = \frac{2\hat{i} - 6\hat{j} + 7\hat{k}}{\sqrt{89}} = \frac{2}{\sqrt{89}}\hat{i} - \frac{6}{\sqrt{89}}\hat{j} + \frac{7}{\sqrt{89}}\hat{k}$$

Example

Show that $\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{C} = 3\hat{i} + \hat{j} - \hat{k}$ are coplanar.

Solution

The necessary and sufficient condition for three vectors $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$, $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$ and $\vec{C} = C_x\hat{i} + C_y\hat{j} + C_z\hat{k}$ to be coplanar is

$$[\vec{A} \ \vec{B} \ \vec{C}] = 0 \quad \text{or} \quad \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = 0$$

$$\text{Now } [\vec{A} \ \vec{B} \ \vec{C}] = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 & -3 \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{vmatrix}$$

$$= 0$$

Therefore, the vectors $\vec{A}, \vec{B}, \vec{C}$ are coplanar.

Example

If $\vec{A} \cdot \vec{B} = \sqrt{3}$ and $\vec{A} \times \vec{B} = \hat{i} + 2\hat{j} + 2\hat{k}$, find the angle between $\vec{A}$ and $\vec{B}$.

Solution

Given

Example

Find all vectors $\vec{V}$ such that $(\hat{i} + 2\hat{j} + \hat{k}) \times \vec{V} = 3\hat{i} + \hat{j} - 5\hat{k}$.

Solution

Given

Differentiation of Vectors: velocity and acceleration

Example

A particle moves along a curve whose parametric equations are $x = e^{-t}$, $y = 2 \cos 3t$, $z = 2 \sin 3t$ where t is the time.

- (a) Determine its velocity and acceleration at any time
- (b) Find the magnitudes of the velocity and acceleration at $t = 0$.

Solution

We perform some

Gradient of a scalar field

Gradient measures the rate of change in a scalar field; the gradient of a scalar field is a vector field. The derivative/differentiation/rate of change of a scalar field result in a vector field called the gradient. It computes the gradient of a scalar function. That is, it finds the Gradient, the slope, how fast you change, in any given direction. A gradient is applied to a scalar quantity that is a function of a 3D vector field: position. The gradient measures the direction in which the scalar quantity changes the most, as well as the rate of change with respect to position. A common example of this is height as a function of latitude and longitude, often applied to mountain ranges. A measure of the slope, and direction of the slope, is often called the gradient.

The vector differential operator is $\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$.

If $\phi(x, y, z)$ is a scalar function, then the gradient of ϕ is defined as

$$\text{grad}(\phi) = \vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$$

Note that, while ϕ is a scalar function, $\text{grad} \phi$ is a vector function.

Example

If $\varphi(x, y, z) = 3x^2y - y^3z^2$, find the gradient of φ or $\text{grad}(\varphi)$ or $\vec{\nabla}\phi$ at the point $(1, -2, -1)$.

Solution

$$\begin{aligned}\text{Grad}(\phi) &= \vec{\nabla}\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \phi \\&= \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k} \\&= \frac{\partial}{\partial x}(3x^2y - y^3z^2)\hat{i} + \frac{\partial}{\partial y}(3x^2y - y^3z^2)\hat{j} + \frac{\partial}{\partial z}(3x^2y - y^3z^2)\hat{k} \\&= (6xy - 0)\hat{i} + (3x^2 - 3y^2z^2)\hat{j} + (0 - 2y^3z)\hat{k} \\&= 6xy\hat{i} + (3x^2 - 3y^2z^2)\hat{j} - 2y^3z\hat{k}\end{aligned}$$

at the point $(1, -2, -1)$, the gradient is

$$\begin{aligned}\text{Grad}(\phi) &= 6 \cdot 1 \cdot (-2)\hat{i} + (3 \cdot 1^2 - 3(-2)^2 \cdot (-1)^2)\hat{j} - 2 \cdot (-2)^3 \cdot (-1)\hat{k} \\&= -12\hat{i} - 9\hat{j} - 16\hat{k}\end{aligned}$$

Example

Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$.

Solution

A normal to the surface $x^2y + 2xz = 4$ is

$$\begin{aligned}\vec{\nabla}\phi &= \vec{\nabla}(x^2y + 2xz - 4) \\&= \frac{\partial}{\partial x}(x^2y + 2xz - 4)\hat{i} + \frac{\partial}{\partial y}(x^2y + 2xz - 4)\hat{j} + \frac{\partial}{\partial z}(x^2y + 2xz - 4)\hat{k} \\&= (2xy + 2z)\hat{i} + x^2\hat{j} + 2x\hat{k}\end{aligned}$$

Thus, the normal at the point $(2, -2, 3)$ is

$$\vec{\nabla}\phi = -2\hat{i} + 4\hat{j} + 4\hat{k}$$

Therefore, the unit normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$ is obtained by

$$\hat{n} = \frac{\vec{\nabla}\phi}{|\vec{\nabla}\phi|} = \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{|-2\hat{i} + 4\hat{j} + 4\hat{k}|} = \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{\sqrt{(-2)^2 + 4^2 + 4^2}} = \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{6} = -\frac{1}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$$

Example

Find the directional derivative of the function $\phi = x^2z + 2xy^2 + yz^2$ at the point $(1, 2, -1)$ in the direction of the vector $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$.

Solution

We start with finding the gradient

$$\begin{aligned}\vec{\nabla}\phi &= \vec{\nabla}(x^2z + 2xy^2 + yz^2) \\ &= \frac{\partial}{\partial x}(x^2z + 2xy^2 + yz^2)\hat{i} + \frac{\partial}{\partial y}(x^2z + 2xy^2 + yz^2)\hat{j} + \frac{\partial}{\partial z}(x^2z + 2xy^2 + yz^2)\hat{k} \\ &= (2xz + 2y^2)\hat{i} + (4xy + z^2)\hat{j} + (x^2 + 2yz)\hat{k}\end{aligned}$$

at the point $(1, 2, -1)$

$$\vec{\nabla}\phi = 6\hat{i} + 9\hat{j} - 3\hat{k}$$

Then, the directional derivative in the direction of the vector $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$ will be

$$\begin{aligned}\hat{a} \cdot \vec{\nabla}\phi &= \frac{\vec{A}}{|\vec{A}|} \cdot \vec{\nabla}\phi \\ &= \left(\frac{2\hat{i} + 3\hat{j} + \hat{k}}{\sqrt{2^2 + 3^2 + 1^2}} \right) \cdot (6\hat{i} + 9\hat{j} - 3\hat{k}) \\ &= \left(\frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k} \right) \cdot (6\hat{i} + 9\hat{j} - 3\hat{k}) \\ &= \frac{36}{\sqrt{14}}\end{aligned}$$

Divergence of a vector field

Divergence measures a vector field's tendency to originate from or convergent upon a given point. It computes the divergence of a vector function. That is, it finds how much "stuff" is leaving a point in space. A divergence is applied to a vector as a function of position, yielding a scalar. The divergence actually measures how much the vector function is spreading out. If you are at a location from which the vector field tends to point away in all directions, you will definitely have a positive divergence. If the field points inward toward a point, the divergence in and near that point is negative. If just as much of the vector field points in as out, the divergence will be approximately zero.

If we form the scalar (dot) product of $\vec{\nabla}$ with a vector function

$$\vec{A}(x, y, z) = A_x(x, y, z)\hat{i} + A_y(x, y, z)\hat{j} + A_z(x, y, z)\hat{k}$$

we get a scalar result called the divergence of $\vec{A}$.

$$\text{div}\vec{A} = \vec{\nabla} \cdot \vec{A} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

If the divergence of a vector field function is zero i.e. $\text{div}\vec{A} = \vec{\nabla} \cdot \vec{A} = 0$, the vector field function $\vec{A}$ is said to be solenoidal.

Example

If $\vec{V}(x, y, z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$, find the divergence of $\vec{V}$ or $\text{div}\vec{V}$ or $\vec{\nabla} \cdot \vec{V}$.

Solution

Given vector field is $\vec{V}(x, y, z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$

The divergence of $\vec{V}$ is given by

$$\begin{aligned} \vec{\nabla} \cdot \vec{V} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (xz\hat{i} + xyz\hat{j} - y^2\hat{k}) \\ &= \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(xyz) + \frac{\partial}{\partial z}(-y^2) \\ &= z + xz \end{aligned}$$

Curl of a vector field

In vector calculus, the curl (or rotor) is a vector operator that describes the infinitesimal rotation of a 3-dimensional vector field. At every point in the field, the curl is represented by a vector. The attributes of this vector (length and direction) characterize the rotation at that point. The direction of the curl is the axis of rotation, as determined by the right-hand rule, and the magnitude of the curl is the magnitude of rotation. If the vector field represents the flow velocity of a moving fluid, then the curl is the circulation density of the fluid. A vector field whose curl is zero is called irrotational. The curl is a form of differentiation for vector fields. Curl measures a vector field's tendency to rotate about a point; the curl of a vector field is another vector field. It computes the rotational aspects of a vector function, maybe people thought how vectors "curl" around a center point, like wind curling around a low pressure on a weather map. A curl measures just that, the curl of a vector field. Unlike the divergence, a curl yields a vector. A vector field that tends to point around an axis, such as vectors pointing tangential to a circle, will yield a non-zero curl with the axis around which the curl occurs as the direction. Another example is the velocity field of motion spiraling in or out, such as a whirlpool. Point your right-hand thumb along the direction of the curl. Curl your fingers around this axis. They will curl in the same direction as the vector field. I do not know the names of the texts, but I know there are books available with vector fields to illustrate both divergence and curl.

The curl of a vector field $\vec{A}(x, y, z) = A_x(x, y, z)\hat{i} + A_y(x, y, z)\hat{j} + A_z(x, y, z)\hat{k}$ is defined by

$$\text{curl}\vec{A} = \vec{\nabla} \times \vec{A} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \times (A_x\hat{i} + A_y\hat{j} + A_z\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

A vector field $\vec{A}$ for which the curl vanishes i.e. $\text{curl}\vec{A} = \vec{\nabla} \times \vec{A} = 0$ is said to be irrotational field or conservative vector field.

Example

If $\vec{V}(x, y, z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$, find the curl of $\vec{V}$ or $\text{curl}\vec{V}$ or $\vec{\nabla} \times \vec{V}$.

Solution

Given vector field is $\vec{V}(x, y, z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$

The curl of $\vec{V}$ is given by

$$\begin{aligned} \vec{\nabla} \times \vec{V} &= \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \times (xz\hat{i} + xyz\hat{j} - y^2\hat{k}) \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xyz & -y^2 \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y}(-y^2) - \frac{\partial}{\partial z}(xyz) \right] - \hat{j} \left[\frac{\partial}{\partial x}(-y^2) - \frac{\partial}{\partial z}(xz) \right] + \hat{k} \left[\frac{\partial}{\partial x}(xyz) - \frac{\partial}{\partial y}(xz) \right] \\ &= -(2y + xy)\hat{i} + x\hat{j} + yz\hat{k} \end{aligned}$$

Example

If $\vec{A}(x, y, z) = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$, find $\vec{\nabla} \times \vec{A}$ (or curl A) at the point $(1, -1, 1)$.

Solution

Given vector field is $\vec{A}(x, y, z) = xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k}$

The curl of $\vec{A}$ is given by

$$\vec{\nabla} \times \vec{A} = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) \times (xz^3\hat{i} - 2x^2yz\hat{j} + 2yz^4\hat{k})$$

$$\begin{aligned}
&= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix} \\
&= \hat{i} \left[\frac{\partial}{\partial y} (2yz^4) - \frac{\partial}{\partial z} (-2x^2yz) \right] - \hat{j} \left[\frac{\partial}{\partial x} (2yz^4) - \frac{\partial}{\partial z} (xz^3) \right] + \hat{k} \left[\frac{\partial}{\partial x} (-2x^2yz) - \frac{\partial}{\partial y} (xz^3) \right] \\
&= (2z^4 + 2x^2y)\hat{i} - 3xz^2\hat{j} - 4xyz\hat{k}
\end{aligned}$$

at the point $(1, -1, 1)$

$$\vec{\nabla} \times \vec{A} = 3\hat{j} + 4\hat{k}$$

Solenoidal and Irrotational vector field

If any vector field has the divergence zero, it is said to be solenoidal. That is, for a solenoidal vector field $\vec{\nabla} \cdot \vec{A} = 0$.

If the curl of any vector field vanishes, it is said to be irrotational or conservative vector field. That is, for an irrotational or conservative vector field $\vec{\nabla} \times \vec{A} = 0$.

Example

Determine the force field $\vec{F}(x, y, z) = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$ is a conservative vector field or not.

Solution

All that we need to do is compute the curl and see if we get the zero vector or not.

The given force field is

$$\vec{F}(x, y, z) = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$$

$$\begin{aligned}
\text{Now, curl of } \vec{F} \text{ is } \vec{\nabla} \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & xyz & -x^2y^2 \end{vmatrix} \\
&= (-2x^2y - xy)\hat{i} + 2xy^2\hat{j} + (yz - x^2)\hat{k} \\
&\neq 0
\end{aligned}$$

The curl isn't the zero vectors and so this vector field is not conservative.

Example

If $\varphi(x, y, z)$ is any solution of the Laplace equation, then show that $\vec{\nabla}\varphi$ is a vector both solenoidal and irrotational.

Solution

For a solenoidal vector field $\vec{\nabla} \cdot \vec{\nabla}\varphi = 0$.

For an irrotational or conservative vector field $\vec{\nabla} \times \vec{\nabla}\varphi = 0$.

Since $\varphi(x, y, z)$ is a solution of the Laplace equation, we have

$$\nabla^2 \varphi = 0$$

which implies that

$$\vec{\nabla} \cdot \vec{\nabla}\varphi = 0$$

Therefore, the vector $\vec{\nabla}\varphi$ is solenoidal.

$$\begin{aligned} \text{Again } \vec{\nabla} \times \vec{\nabla}\varphi &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \varphi}{\partial x} & \frac{\partial \varphi}{\partial y} & \frac{\partial \varphi}{\partial z} \end{vmatrix} \\ &= \left(\frac{\partial^2 \varphi}{\partial y \partial z} - \frac{\partial^2 \varphi}{\partial z \partial y} \right) \hat{i} + \left(\frac{\partial^2 \varphi}{\partial x \partial z} - \frac{\partial^2 \varphi}{\partial z \partial x} \right) \hat{j} + \left(\frac{\partial^2 \varphi}{\partial x \partial y} - \frac{\partial^2 \varphi}{\partial y \partial x} \right) \hat{k} \\ &= 0 \cdot \hat{i} + 0 \cdot \hat{j} + 0 \cdot \hat{k} \\ &= 0 \end{aligned}$$

Hence, the vector $\vec{\nabla}\varphi$ is irrotational.

Line Integral

Any Integral which is evaluated along the curve is called Line Integral, and it is denoted by

$\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F}$ is a vector point function, $\vec{r}$ is a position vector and C is the curve.

The work performed by a vector field on a particle moving along a parametric curve C is obtained by integrating the scalar tangential component of force along C .

$$W = \oint_C \vec{F} \cdot d\vec{r}$$

If $\vec{F}(x, y) = P(x, y)\hat{i} + Q(x, y)\hat{j}$ and $d\vec{r} = dx\hat{i} + dy\hat{j}$, then

$$\vec{F} \cdot d\vec{r} = P(x,y)dx + Q(x,y)dy$$

Thus, the workdone = $\oint_c \vec{F} \cdot d\vec{r} = \oint_c Pdx + Qdy$

In three dimension,

$$\text{workdone} = \oint_c \vec{F} \cdot d\vec{r} = \oint_c Pdx + Qdy + Rdz$$

Example

How much work is accomplished by the force $\vec{F}(x,y) = xy\hat{i} + y\hat{j}$ in pushing a particle from (0,0) to (3,9) along the parabola $y = x^2$?

Solution

$$\vec{F}(x,y) = xy\hat{i} + y\hat{j}$$

$$\text{Workdone} = \oint_c \vec{F} \cdot d\vec{r}$$

$$= \oint_c Pdx + Qdy$$

$$= \oint_c xy \, dx + y \, dy$$

The path C is said to be the parabola $y = x^2$, giving $dy = 2x \, dx$

$$\text{Workdone} = \oint_c xy \, dx + y \, dy$$

$$= \oint_0^3 x \cdot x^2 \, dx + x^2 \cdot 2x \, dx$$

$$= \oint_0^3 3x^3 \, dx$$

$$= \frac{243}{4} \text{ unit.}$$

Example

Evaluate $\oint_c xy \, dx$ from B(1,0) to C(0,1) along the curve C that is the portion of the unit circle $x^2 + y^2 = 1$ in the first quadrant.

Solution

$$x^2 + y^2 = 1$$

$$y = \sqrt{1 - x^2}$$

The curve BC is the first quadrant of the unit circle as shown in figure. On the curve $y = \sqrt{1 - x^2}$, so that,

$$\oint_c xy \, dx = \int_1^0 x\sqrt{1 - x^2} \, dx$$